

TARGET-SPECIFIC RECOVERABILITY MAPS FOR REDUCED-LEAD ECG RECONSTRUCTION

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ABSTRACT

Reconstructing missing electrocardiogram (ECG) leads from a reduced set is ill-posed, and a scalar reconstruction error says nothing about *which* named feature, on *which* lead, a clinician can trust. We give a *target-specific recoverability map*: for a waveform segment s (P/QRS/ST/T), an observed lead set S , and a target lead ℓ , two closed-form numbers derived from one truncated SVD of the segment’s empirical rank-3 spatial (“dipolar”) sub-matrix—an *identifiability* $\eta_{s,\ell}(S)$ (zero iff the target’s low-rank component is recoverable from S) and a *conditioning* $\kappa_{s,\ell}(S)$ (its noise/observed-residual gain). On PTB-XL the map is stable to the truncation tolerance for the well-posed configurations and rank-sensitive (with reported sensitivity) for the ill-posed ones; e.g. it proves *a priori* that precordial ST is not identifiable from limb leads. We add distribution-free calibrated intervals for the empirically predictable residual (conformalized quantile regression with strict train/ calibrate/test fold discipline; within-group coverage near nominal), and contextualize with reconstruction baselines. The map is reconstructor-agnostic and turns “trust the reconstruction” into a per-feature, per-lead, calibrated statement.

Index Terms— electrocardiography, inverse problems, identifiability, conformal prediction, trustworthy machine learning

1. INTRODUCTION

Wearable and reduced-lead ECG acquisition makes lead *reconstruction* ubiquitous: recover the standard 12 leads from one, three, or six observed leads [1, 2]. A low aggregate error certifies nothing about *which* morphological features are trustworthy: a cardiologist reads the P wave, the QRS complex, the ST segment and the T wave, each on specific leads.

We ask a sharper, reconstructor-independent question: *for a given observed lead set, which target lead’s dipolar morphology is even identifiable, and how well-conditioned is its recovery?* Each waveform segment’s instantaneous potential is approximately low rank across leads; we estimate a per-segment rank-3 spatial basis M_s (top-3 singular vectors of the population lead covariance). We stress that M_s is a *population-estimated* (PCA) object: on real PTB-XL it shares

two directions with the classical vectorcardiographic (inverse-Dower) dipole but its third direction differs materially, so we call it an empirical rank-3 subspace rather than a physical dipole (Sec. 3).

Contributions. (1) A *target-specific* recoverability map: closed-form per-lead identifiability $\eta_{s,\ell}(S)$ and conditioning $\kappa_{s,\ell}(S)$ from a single truncated SVD, replacing coarse “global rank<3” rules, with a truncation-tolerance sensitivity analysis and record-bootstrap confidence intervals (Sec. ??). (2) A distribution-free calibration layer for the empirically predictable residual, with strict fold discipline and honestly stated within-group marginal coverage (Sec. 2). (3) Evidence on PTB-XL: reconstruction baselines that quantify what each configuration can recover, and a certificate-level safety statement—precordial ST is not identifiable from limb leads—available before any reconstruction (Sec. 3).

Model. Within segment s write the population 12-lead vector as $L_s = \mu_s + M_s d + r_s$, where $M_s \in \mathbb{R}^{12 \times 3}$ is the (population-estimated, PCA) dipolar basis, d the dipole coordinates, and r_s the non-dipolar residual. Observing S gives $y_S = \text{Sel}_S L_s + n$. Let $M_{s,S} = \text{Sel}_S M_s$ have truncated SVD at relative tolerance g ; $M_{s,S}^+$ its truncated pseudo-inverse and $P_{\text{obs}} = M_{s,S}^+ M_{s,S}$ the projector onto the observed dipole coordinate directions.

Theorem 1 (Per-target-lead dipolar identifiability and conditioning). *The mean-centred dipolar estimate $\hat{L}_s = \mu_s + M_s M_{s,S}^+ (y_S - \mu_{s,S})$ satisfies, for each target lead ℓ with e_ℓ the ℓ -th unit vector:*

- (i) *Identifiability. Define $\eta_{s,\ell}(S) = \|e_\ell^\top M_s (I - P_{\text{obs}})\|_2$. If $\eta_{s,\ell}(S) = 0$, the dipolar component of lead ℓ is identifiable from S : $e_\ell^\top M_s d$ is determined by $\text{Sel}_S M_s d$. If $\eta_{s,\ell}(S) > 0$, there is a dipole direction unobserved by S that changes lead ℓ ; its dipolar component is not identifiable at any SNR.*
- (ii) *Conditioning. On the identifiable part, the error from observation noise and from the observed non-dipolar residual is amplified into lead ℓ by $\kappa_{s,\ell}(S) = \|e_\ell^\top M_s M_{s,S}^+\|_2$: $|e_\ell^\top (\hat{L}_s - \mu_s - M_s P_{\text{obs}} d)| \leq \kappa_{s,\ell}(S) (\|\text{Sel}_S r_s\| + \|n\|)$.*

The global $\kappa_s(S) = \|M_s M_{s,S}^+\|_2 = \max_\ell \kappa_{s,\ell}(S)$ is a configuration-level worst case. Both $\eta_{s,\ell}$ and $\kappa_{s,\ell}$ depend

on the estimated M_s and on ϱ ; near-rank-deficient S are reported with an ϱ -sensitivity sweep and record-bootstrap confidence intervals.

The non-dipolar residual (no independence claim on real data). Let $m_s(y_S) = \mathbb{E}[r_s | y_S]$ and $u_s = r_s - m_s(y_S)$, so $\mathbb{E}[u_s | y_S] = \mathbf{0}$. We do *not* assume $u_s \perp y_S$. The Bayes-optimal reconstruction error of the residual is

$$\inf_{\hat{r}} \mathbb{E} \|r_s - \hat{r}(y_S)\|^2 = \mathbb{E} [\text{Var}(r_s | y_S)].$$

Part of r_s (the *empirically predictable residual*) is recovered by a predictor trained on y_S and metered with distribution-free calibrated intervals; the remainder is an *achievability gap*, established by a held-out achievability analysis for a stated predictor family—not declared unrecoverable a priori.

Proposition 1 (Strict independence limit – synthetic model only). *If the data are generated so that a component u is statistically independent of y_S (an explicit synthetic construction), then $p(u | y_S) = p(u)$, the Bayes estimate of u is its prior mean, and $\mathbb{E} \|\hat{u} - u\|^2 \geq \text{Var}(u)$ for any estimator. This strict limit is invoked only for the synthetic model; on real ECG we make no such independence claim.*

2. DISTRIBUTION-FREE CALIBRATION

Predictable-residual intervals. For each pre-registered group $g = (S, s, \ell)$ we train a quantile regressor of the target lead’s off-dipole residual on the observed leads (gradient-boosted pinball loss) and apply conformalized quantile regression [3] with a Mondrian (per-group) correction. Fold discipline is strict and leakage-free: basis and quantile model on folds 1–7, conformal calibration on fold 9, a single evaluation on fold 10. We report *within-group marginal coverage* under exchangeability (not per-example conditional coverage), interval width, and group sizes.

Off-dipole flag. We threshold the off-dipole energy at the one-sided $(1 - \alpha)$ conformal quantile over faithful reconstructions; the false-flag rate is $\leq \alpha$ for a fresh reconstruction *exchangeable with the calibration distribution*—not distribution-free under an arbitrary shift.

3. EXPERIMENTS

Data. PTB-XL [4] (21,799 twelve-lead records, 100/500 Hz, 10-fold split) with P/QRS/ST/T delineation by NeuroKit2 [5].

3.1. The recoverability map on PTB-XL

We estimate M_s (rank 3) per segment on the training folds and evaluate the map on 1,200 NORM records (per-lead η, κ with 200-sample bootstrap CIs, truncation tolerance $\varrho=10^{-2}$; Fig. 1). The map is per target lead: a single observed lead (e.g.

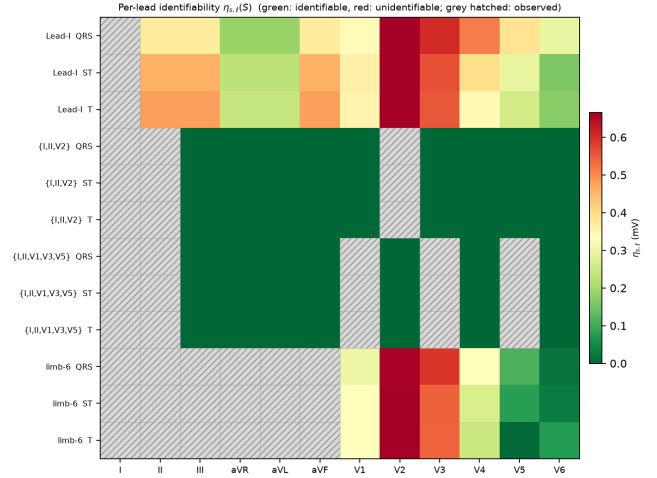


Fig. 1. Per-lead identifiability $\eta_{s,\ell}(S)$ across configurations and segments (green: identifiable, red: unidentifiable; grey hatched: observed). limb-6 leaves all precordial leads unidentifiable.

Lead I) leaves *every* other lead’s dipolar component unidentifiable ($\eta > 0$; rank 1); a dipole-spanning triplet $\{I, II, V2\}$ or the five-lead $\{I, II, V1, V3, V5\}$ makes all targets identifiable ($\eta \approx 0$). For the six limb leads, *all* precordial leads V1–V6 are unidentifiable for QRS/ST/T: the transverse dipole is unobserved, so precordial ST cannot be read from limb leads—a warning available before any reconstruction.

Truncation-tolerance sensitivity. Rank/ κ of well-posed configurations are stable across $\varrho \in \{10^{-4}, \dots, 10^{-1}\}$ (e.g. $\{I, II, V1, V3, V5\}$ is rank 3 throughout), while near-rank-deficient sets are sensitive: on synthetic data $\{V1, V2, V3\}$ is rank 2 at $\varrho=10^{-2}$ but rank 3 with $\kappa \sim 200$ at $\varrho=10^{-4}$. We therefore report a single κ only for well-conditioned configurations and give the ϱ -sweep and bootstrap CIs otherwise; a single 10^4 or 10^5 figure for a degenerate set is not meaningful.

3.2. Calibrated intervals for the predictable residual

For the empirically predictable residual we train a gradient-boosted quantile regressor of each target lead’s off-dipole residual on the observed leads and apply Mondrian CQR per (S, s, ℓ) group, with strict fold discipline (basis and model on folds 1–7, calibration on fold 9, a single evaluation on fold 10; nominal $1 - \alpha = 0.90$). Across the pre-registered groups of $\{I, II, V1, V3, V5\}$ and $\{I, II, V2\}$, fold-10 within-group coverage clusters near nominal (0.88–0.95; e.g. $\{I, II, V1, V3, V5\}$ QRS/V2 0.92, record-bootstrap CI [0.89, 0.95], mean width 0.165 mV), with a few groups slightly below—the expected finite-sample behaviour of within-group marginal coverage. This certifies *what interval* the predictable residual admits, per feature and lead, without any distribution-free claim be-

Table 1. Reconstruction RMSE (mV), target leads (QRS). “dip” = Tier-I dipolar; “ridge” = per-segment ridge (Tier I+II).

config	prior-mean	inv-Dower	dip	ridge
{I,II,V1,V3,V5}	0.111	0.063	0.053	0.039
limb-6	0.135	0.302	0.117	0.117

yond exchangeability with the calibration fold.

3.3. Baselines and what the map predicts

Table 1 reports reconstruction RMSE (mV) of the unobserved leads for classical baselines (3,000 train / 1,500 test, record-bootstrap CIs). On a dipole-spanning set the learned linear ridge/OLS (dipolar plus predictable residual) beats the pure dipolar recovery, which beats inverse-Dower and the prior mean—there is recoverable non-dipolar content, exactly the calibrated Tier II layer. On limb-6 *every* reconstructor barely beats the prior mean (QRS 0.117 vs 0.135; ST 0.055 vs 0.060 mV) and inverse-Dower is *worse* than the prior mean (0.30): the map’s $\eta > 0$ warning is borne out—limb→precordial is near-unrecoverable, so a low reconstruction error there would be spurious.

Empirical, not physical, dipole. On real PTB-XL the estimated M_s shares two directions with the classical inverse-Dower vectorcardiographic space (principal angles 2–6°) but its third direction differs materially (max angle 43–55°, bootstrap CIs excluding small angles). We therefore treat M_s as an *empirical rank-3 spatial subspace*, not a physical cardiac dipole; the earlier synthetic “matches inverse-Dower” check used inverse-Dower to generate the data and is not evidence on real ECG. All certificate quantities are stated with respect to this estimated subspace.

4. CONCLUSION

Reduced-lead ECG reconstruction has a *known low-rank structure per segment*. Exploiting it gives a per-target-lead recoverability map—what is identifiable, how well-conditioned, and (via calibration) what interval the predictable residual admits—before and independent of any reconstructor. It tells a clinician which lead configuration can support a decision, e.g. that precordial ST cannot be read from limb leads.

5. REFERENCES

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